

KAGGLE · BUILD WITH AI — GEMMA HACKATHON · GDG WINDSOR · EDGE / ON-DEVICE TRACK

# GEMMA MONSTERS

A monster-battling castle with a real tutoring agent inside.

Five monsters guard the Citadel. Each one is a trick — a wrong idea that feels right.

Powered by Gemma, running entirely on one laptop. Nothing a student does ever leaves it.

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## Wrong answers are not random

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$$\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$$

Tops with tops, bottoms with bottoms. Feels tidy. Completely wrong.

**That is a trick: the wrong idea that feels right.**

This student does not need more practice — they need that one idea found and removed. Practice apps mark the answer wrong and move on. Cloud tutors need the internet, ship a child's work to a server, and sometimes invent mathematics.

# So we turned every trick into a monster

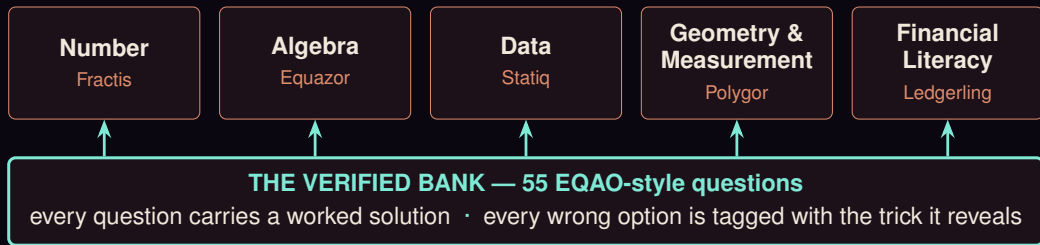
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| Monster    | Guards the strand      | The trick it plants in your head                     |
|------------|------------------------|------------------------------------------------------|
| Fractis    | Number                 | “just add fractions straight across”                 |
| Equazor    | Algebra                | signs that flip the wrong way across the equals sign |
| Statiq     | Data                   | mean and median blurred into one fuzzy word          |
| Polygor    | Geometry & Measurement | stolen formulas that <i>almost</i> fit               |
| Ledgerling | Financial Literacy     | interest quietly skimmed off a sloppy budget         |

You cannot beat a monster by luck. You beat it by proving its trick no longer fools you.

# Built on the real Grade 9 curriculum

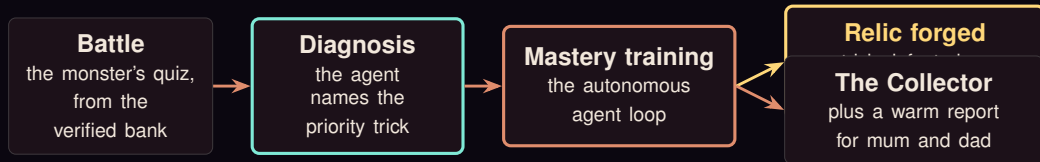
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So when a student picks a wrong answer, **the wrong answer itself is the diagnosis** — a table lookup, not a model guess.

# One session, start to finish

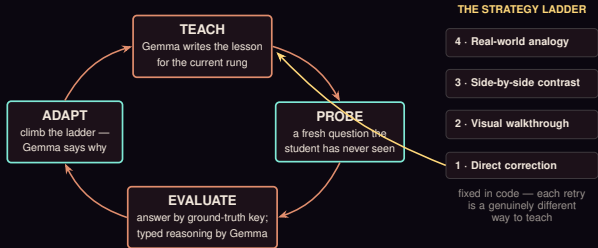
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- The monster meets you **by name** — and it **remembers**: Gemma writes its greeting from your real battle history.
- Between battles: **90-second mental-math skirmishes** against the Collector's lieutenants, and his three-life **boss speed trial**.
- Defeat all five monsters → the golden gate opens → **the rescue finale**.

# The agent: it keeps teaching until the trick is gone

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**Teal = code decides. Coral = Gemma judges.**

**Deterministic spine:** plain code runs the loop; Gemma is called only at the judgment points.

**Two exits, no third:** two fresh correct answers with solid reasoning **forges a relic**; a spent ladder means a **warm report for mum and dad**. The agent never gives up silently.

Every switch of strategy is announced on screen, with its

# A right answer is not enough

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## THE PROBE

Calculate  $\frac{1}{2} + \frac{1}{3}$  — picks  $\frac{5}{6}$  (correct)

## THE STUDENT TYPES

*"I just guessed the biggest fraction, that usually works"*

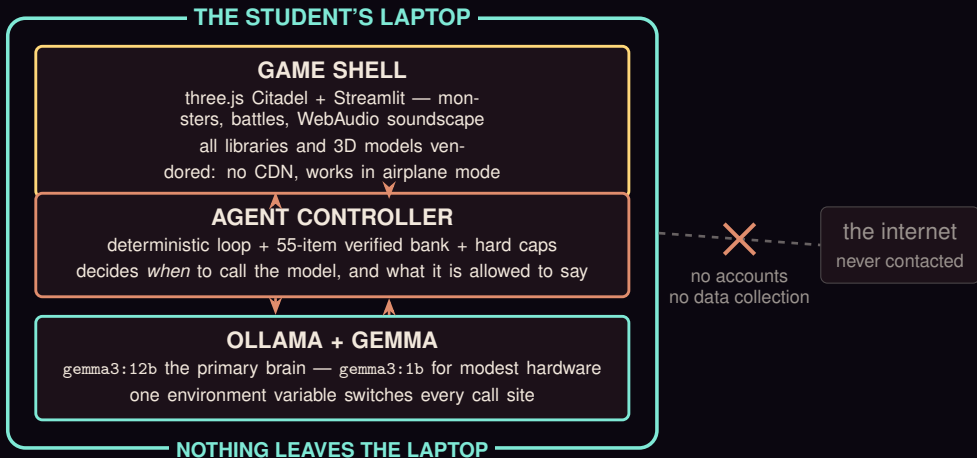
## GEMMA'S GRADE

**SHALLOW** — does **not** count toward mastery

- Three labels only — **RESOLVED / SHALLOW / SAME\_ERROR**. A closed set, nothing open-ended.
- The monster **answers the student's own words back, in character** — it read what you wrote.
- **Fail-open**: if the grader hiccups, the student is never punished for it.

A lucky bubble cannot fake understanding past this gate.

# Everything on one laptop



The Edge / On-Device promise, kept literally: **a student's struggles stay** 8



# Gemma judges. Code guarantees.

## WHAT GEMMA DECIDES

the judgment calls

writes every lesson and study guide  
grades the student's typed reasoning  
picks the next ladder rung — and says why  
authors fresh probes when the bank runs dry  
speaks as the monsters; forges the relics  
phrases the warm report for mum and dad

## WHAT CODE GUARANTEES

the ground truth

every math fact comes from the verified bank  
answers graded against the ground-truth key  
the trick is diagnosed by tag lookup  
the ladder's order is fixed  
hard caps end every session  
scores and history computed, then injected

**THE GROUNDING RULE** — the verified worked solution rides inside every prompt:

Gemma explains *why*; it never recomputes the math and never states new numbers.

The model does the thinking — **it is never the source of mathematical**

# We assumed the model would fail — and built for it

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## Self-checked fresh probes

The small model once generated a question with a wrong answer key. Now: bank first; a generated probe is used only if Gemma can **solve it blind** and agree with its own key. Failures are discarded.

## No invented arithmetic

The 1B once produced wrong math mid-explanation. Now the model never computes at all: the verified solution is the **only math it may state**.

## Hard caps — it always ends

4 probe cycles, a 12-model-call budget, and ladder exhaustion each force a clean hand-off. **The loop terminates no matter what the model does.**

## The warm hand-off

When teaching stalls, mum and dad get a **warm report**: what the trick is, what was tried, and kitchen-table activities to try together. The agent hands the student to a human — never gives up silently.

# Scored against the rubric

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## USE OF GEMMA 30

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Ten call sites, one auditable door — lessons, reasoning grades, strategy picks, probes, coaching, relics, reports.

Every site guardrailed; grounded in the verified bank.

Fully local via Ollama: `gemma3:12b`, with 1b one flag away.

## INNOVATION 30

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A tutoring agent you can *fight*: each trick embodied as a monster.

Mastery gated on the student's own typed reasoning — and the monster answers those words back.

A game world that runs on judgment, with a deterministic spine.

## FUNCTIONALITY 20

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Runs live, end to end, Wi-Fi off: Citadel, battles, agent loop, skirmishes, boss speed trial, rescue finale.

Hard caps mean no session can hang or loop forever.

Fail-open grading: a model hiccup never hurts the student.

## WRITEUP 20

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An honest engineering story: what the model broke in testing, and the structural fix for each failure.

Architecture doc, per-call-site guardrail table, and a companion Kaggle notebook that reproduces the agent loop.

# Every monster is a trick — a wrong idea that feels right.

Defeat the idea, and the gate opens.

A student walks in adding fractions straight across. Twenty minutes later they have *proven* the trick is gone — and nothing about them ever left their laptop.

[github.com/EdTechDL/gemma-without-borders](https://github.com/EdTechDL/gemma-without-borders) · live demo, Wi-Fi off · companion  
Kaggle notebook

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